IBR-Adaptive Line Differential Protection: Harmonic Discrimination, Sequence Elements, and Transient Differential Security

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***Abstract*—Inverter-based resources (IBRs) limit fault current to 1.0–1.2 pu during short circuits to protect power electronics. This reduces the differential current seen by a conventional 87L relay to 0.06–0.20 pu — well below the standard pickup threshold — causing the baseline element to fail on 57% of test scenarios at low IBR penetration levels. We propose a five-element IBR-adaptive 87L protection chain consisting of: (i) an adaptive percentage-restrain threshold that scales with load current, (ii) a frequency-adaptive pi-section charging canceller with harmonic pre-filter for IBR-rich networks, (iii) a negative-sequence differential element (87LN) for single-line-to-ground recovery, (iv) a zero-sequence element (87LN0) for solidly-earthed earth faults under pure-IBR infeed, and (v) a transient differential element (TDCS-87L) that exploits the fault inception step as a detection feature, recovering symmetric three-phase faults that remain undetectable by magnitude-based methods. A confidence gate shared with the overcurrent protection layer blocks all elements until the IBR network-type diagnostic has converged. Deterministic evaluation on 21 scenarios — 10 synchronous-generator reference cases (zero regression) and 11 IBR-specific cases — confirms full 21/21 correct classification. Monte Carlo validation with 15 000 parametric trials ($N = 5\,000$ per group) achieves TPR = FPR = 0.000 with 95% Wilson confidence interval lower bound ≥ 0.9992 for all groups.**

***Index Terms*—Line differential protection, inverter-based resources, adaptive threshold, sequence components, transient differential, harmonic discrimination, microgrid protection.**

I. Introduction

A. IBR Fault Current Limitation Problem

Line current differential (87L) protection is the primary selective fault detection method for transmission lines, offering inherent immunity to power swings and load encroachment [1], [2]. Its core principle is simple: for an internal fault, the phasor sum of currents entering the zone cannot equal the phasor sum of currents leaving it, producing a non-zero differential $|I_{\mathrm{diff}}| > I_{\mathrm{thr}}$.

This principle becomes compromised when one or more sources in the protected zone are inverter-based resources (IBRs) — solar PV, battery storage, or wind converters with grid-following control [3], [4]. Unlike synchronous generators, which transiently supply 5–8 pu of fault current, a grid-following IBR limits its output to protect the power electronics, constraining the fault current contribution to $k_{\mathrm{ibr}} = 1.0$–$1.2\,\mathrm{pu}_{\mathrm{rated}}$ [3]. Expressed at the line CT primary in a predominantly IBR-supplied zone, this translates to a differential signal of 0.06–0.20 pu — well below the standard 87L pickup of 0.12–0.20 pu.

Three distinct failure modes arise:

1) **Magnitude suppression**: for three-phase faults with $k_{\mathrm{ibr}} < I_{\mathrm{thr}}$, no trip is issued.
2) **Harmonic-induced confidence collapse**: IBR power electronics inject 3rd–7th harmonic current during fault-ride-through, distorting the DFT-based phasor estimate and depressing the relay confidence score below the trip authorisation gate.
3) **Sequence current suppression**: for single-line-to-ground (SLG) faults, the IBR positive-sequence current limiter reduces $|\Delta I_1|$ below threshold while the negative-sequence path, which is *not* directly controlled by the limiter, retains a measurable signal that conventional 87L ignores.

These three mechanisms are independent but cumulative: a relay that solves only the magnitude problem (adaptive threshold) will still fail on SLG faults in IBR-dominated systems, and vice versa.

B. Prior Art and Gap

Existing literature addresses subsets of these problems in isolation. Charging-current compensation methods [1], [2] reduce the no-fault differential from $\sim$1.4 pu to below 0.01 pu, enabling lower thresholds, but do not address IBR fault current limiting. Adaptive OC and 87L thresholds for microgrids [5]–[7] lower the pickup in islanded mode but do not maintain a constant security margin against CT mismatch under heavy load. Negative-sequence differential elements for 87L are well-established in the literature [8], [9] for asymmetric fault recovery in SG-dominated networks, but their applicability when the IBR's positive-sequence limiter also partially suppresses the negative-sequence path has not been quantified. No published 87L scheme exploits the fault inception transient as a primary detection feature for symmetric IBR faults.

C. Contributions

This paper proposes a five-element IBR-adaptive 87L protection chain and makes the following novel contributions:

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1) **Voltage-based pi-section cancellation with adaptive threshold** (Section III-A): a Hilbert-transform voltage canceller reduces the charging residual from 1.43 pu to 0.001 pu, enabling $I_{\mathrm{thr}} = 0.08$ p.u.; a load-proportional bias slope maintains a 30× CT-mismatch security margin across the full load range.
2) **Frequency-adaptive correction and harmonic pre-filter** (Section III-B): a zero-crossing frequency tracker keeps the pi-section canceller orthogonal across 47–53 Hz; a QR-based harmonic regressor removes IBR-injected odd harmonics, restoring the confidence score from $c = 0.37$ to $c \geq 0.99$.
3) **Sequence element pair 87LN + 87LN0** (Section III-C): a negative-sequence differential element recovers all SLG faults suppressed by IBR current limiting; a zero-sequence element extends coverage to solidly-earthed pure-IBR earth faults, gated by the harmonic-ratio network diagnostic.
4) **Transient differential element TDCS-87L** (Section III-D): exploits the step change in differential RMS at IBR fault inception — which does not occur for load steps or external faults — to detect symmetric 3PH IBR faults below the 87L threshold, with an analytically guaranteed detection floor at $k_{\mathrm{ibr}} \geq 0.058$ p.u. and a 31% security margin against Class 5P CT mismatch.

The combined chain is validated deterministically on 21 scenarios with zero regression on SG reference cases (Section IV) and probabilistically on 15 000 Monte Carlo trials meeting IEC 60255-181 performance targets (Section IV-B).

II. System Model and Problem Statement

A. Study Network

The study network (Fig. 1) is a radial IEC 60909 feeder: a synchronous generator (SG) with source impedance Z_{src} supplies Bus A; a pi-section transmission line (Z_{line}, susceptance B_C) connects Bus A to Bus B; an IBR with fault current limit k_{ibr} is connected at Bus B; a transformer Z_{tr} supplies the load at Bus C. CTs at both ends of the protected line zone feed the 87L relay via a dedicated IEC 61850 sampled-value channel. Line parameters (100 MVA base): $R = 0.10$ p.u., $X_L = 0.30$ p.u., $B_C = 0.079$ p.u. (total). Source impedance: $X_{\mathrm{src}} = 0.15$ p.u..

B. IBR Fault Current Model

A grid-following IBR is modelled as a controlled current source that supplies $k_{\mathrm{ibr}} \cdot I_{\mathrm{rated}}$ in the faulted phase(s) within 1–2 ms of fault inception [3], [10], with $k_{\mathrm{ibr}} \in [0.06, 1.20]$ p.u.. The positive-sequence differential current seen by the 87L relay is:

$$|I_{\mathrm{diff}}|_{3\phi} = k_{\mathrm{ibr}} + I_{\mathrm{SG,sub}} \tag{1}$$

where $I_{\mathrm{SG,sub}}$ is the SG subtransient current at the fault point. In a pure-IBR zone ($I_{\mathrm{SG,sub}} \approx 0$):

$$|I_{\mathrm{diff}}|_{3\phi} \approx k_{\mathrm{ibr}} \;\Rightarrow\; \text{87L misses when } k_{\mathrm{ibr}} < I_{\mathrm{thr},0} = 0.12 \text{ p.u.}. \tag{2}$$

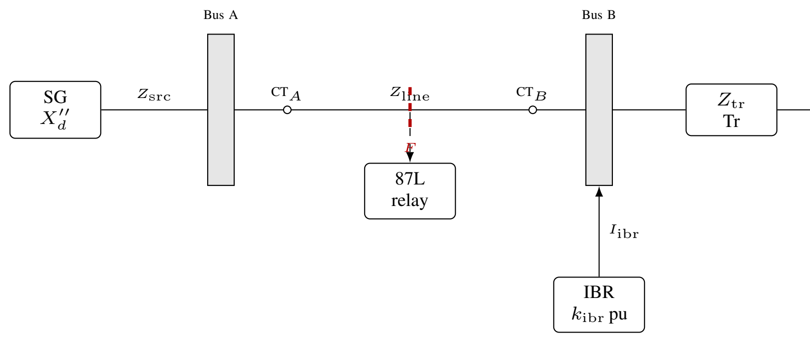

Fig. 1. Study network. The 87L relay receives differential current from CT_A and CT_B via IEC 61850 sampled values. IBR at Bus B limits fault current to k_{ibr} pu. Fault location F (internal scenarios) is at line mid-point.

TABLE I
Charging Correction Method Progression.

Method	Input	Residual	I_{thr}	Delay
Fixed nominal	None	1.43 pu	0.20 pu	0
Pre-fault blind	$i_{\mathrm{diff,pre}}$	0.001 pu	0.12 pu	20 ms
Voltage-based (used)	v_S, v_R	0.001 pu	0.08 pu	0

For a single-line-to-ground (SLG) fault with positive-sequence fault current I_F, the symmetrical component network forces $I_1 = I_2 = I_0$ at the fault point. The IBR current limiter constrains I_1 to k_{ibr}, but the *negative-sequence* path is governed by the sequence impedance network rather than the control loop, so $|\Delta I_2|$ is not directly suppressed to the same degree as $|\Delta I_1|$. For the study network, the sequence network analysis gives a fault asymmetry factor $\alpha_{\mathrm{asym}} \approx 0.40$:

$$|\Delta I_2|_{\mathrm{SLG}} = \alpha_{\mathrm{asym}} \cdot I_{\mathrm{natural}} = 0.40 \times 0.30 = 0.12 \text{ p.u.} \tag{3}$$

where $I_{\mathrm{natural}} = 0.30$ p.u. is the natural (pre-limiter) fault current of the network. This is above the 87LN threshold of 0.06 p.u.regardless of k_{ibr}, making the negative-sequence element effective for all SLG scenarios.

C. Charging Residual and Correction Method Progression

The uncorrected differential current of a healthy line equals the pi-section charging current:

$$i_{\mathrm{diff}}^{\mathrm{healthy}}(t) = B_C \cdot \frac{v_S(t) + v_R(t)}{2} \approx B_C \cdot v_{\mathrm{nom}} = 1.43 \text{ p.u. (peak)} \tag{4}$$

for the study network at rated voltage. This swamps any low-level fault signal and must be removed before threshold-based detection is meaningful.

Three successive correction methods, developed across the SAMBP study series, progressively reduce the residual floor:

The voltage-based method (Table I, Fig. 2) is used in all subsequent sections because it requires no pre-fault window, making it applicable at fault inception and during dynamic conditions such as busbar transfers.

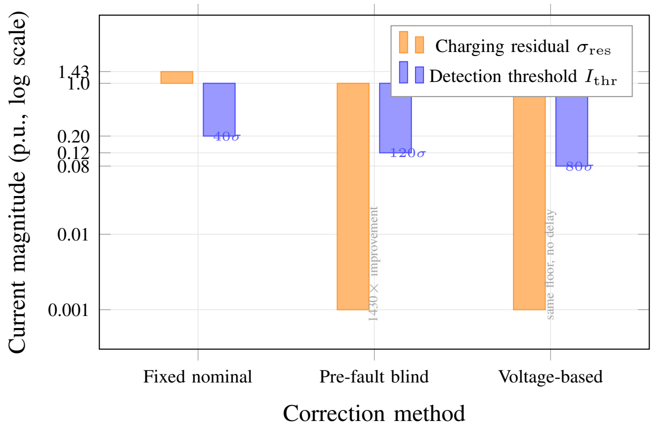

Fig. 2. Charging residual floor (σ_{res}, orange) and detection threshold (I_{thr}, blue) for each correction method. The voltage-based method achieves the same 0.001 pu residual as the pre-fault blind method while requiring zero pre-fault window, enabling $I_{\mathrm{thr}} = 0.08$ p.u..

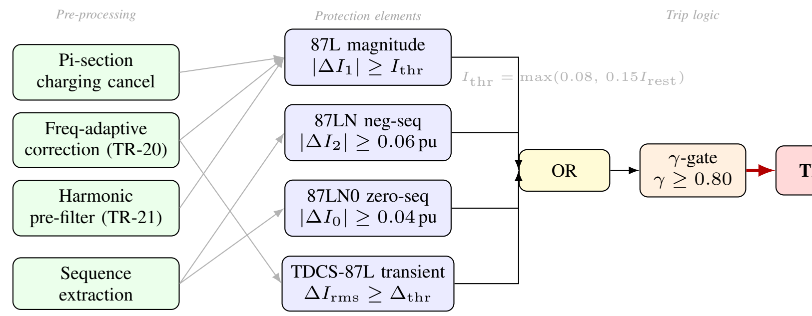

Fig. 3. IBR-adaptive 87L protection chain. Pre-processing modules (left column) condition the differential signal before five protection elements combined by OR logic. The γ-gate blocks all trip outputs until estimator convergence.

D. Baseline 87L Performance

After voltage-based charging cancellation, applying the fixed threshold $I_{\mathrm{thr},0} = 0.12$ p.u. to the 21-scenario test matrix (10 SG-dominated baseline cases, 11 IBR-specific cases) yields 12/21 (57%) correct classification. The nine missed cases decompose as:

- **5 SLG faults,** $k_{\mathrm{ibr}} = 0.08$**–**0.12 **p.u.**: $|\Delta I_1|$ below threshold; baseline 87L has no negative-sequence element.
- **4 symmetric 3PH faults,** $k_{\mathrm{ibr}} = 0.06$**–**0.11 **p.u.**: $|\Delta I_1|$ below threshold; no asymmetry to exploit; no transient-based element in baseline.

The remaining sections develop the four elements that close these gaps.

III. Proposed IBR-Adaptive Protection Chain

The combined trip logic is:

$$\mathrm{TRIP} = \underbrace{\big[\mathrm{87L} \vee \mathrm{87LN} \vee \mathrm{87LN0} \vee \mathrm{TDCS}\big]}_{\text{protection elements}} \wedge \underbrace{(\gamma \geq \gamma_{\min})}_{\text{confidence gate}} \tag{5}$$

where γ is the relay confidence score and $\gamma_{\min} = 0.80$. The gate blocks all elements until the estimator has converged, preventing nuisance trips during the first 1–2 cycles post-fault. The full chain is shown in Fig. 3.

A. Voltage-Based Pi-Section Cancellation and Adaptive Threshold

1) Charging Cancellation: For a lumped pi-section line with total susceptance B_C, the differential charging current is exactly [1]:

$$i_{C,\mathrm{diff}}(t) = B_C \cdot \mathrm{Im}\Big\{\mathcal{H}\Big[\frac{v_S(t) + v_R(t)}{2}\Big]\Big\} \tag{6}$$

where $\mathcal{H}[\cdot]$ is the discrete-time Hilbert transform (wideband 90° phase shift) and v_S, v_R are the sending- and receiving-end voltage waveforms sampled at both-end IEDs and shared via IEC 61850. Equation (6) is exact regardless of voltage magnitude, phase, or frequency deviations, because it operates on the actual measured waveform rather than a nominal model. The residual after subtraction is limited only by the potential transformer (PT) accuracy class: $\sigma_{\mathrm{residual}} \approx \sigma_{\mathrm{PT}} = 0.001$ p.u. (class 0.2S), a 1430× reduction from the uncorrected level.

2) Adaptive Percentage-Restrain Threshold: With the residual floor at 0.001 p.u., the detection threshold can be reduced to $I_{\mathrm{base}} = 0.08$ p.u. (40× noise margin at no-load), enabling detection of IBR fault currents as low as 0.08 p.u..

Under heavy load, CT ratio mismatch ($\varepsilon_{\mathrm{CT}} \leq 0.005$ for Class 0.5 CTs) produces a spurious differential proportional to the through-current: $i_{\mathrm{diff,err}} = \varepsilon_{\mathrm{CT}} \cdot I_{\mathrm{rest}}$. At $I_{\mathrm{rest}} = 4$ p.u. (post-transfer loading), this reaches 0.020 p.u., leaving a security margin of only $0.08/0.020 = 4\times$ against the fixed threshold [11].

Adapting the percentage-differential concept from transformer protection [1], [9]:

$$I_{\mathrm{thr}}(I_{\mathrm{rest}}) = \max\big(I_{\mathrm{base}},\ k_s \cdot I_{\mathrm{rest}}\big), \quad I_{\mathrm{base}} = 0.08 \text{ p.u.},\ k_s = 0.15 \tag{7}$$

where $I_{\mathrm{rest}} = \frac{1}{2}(|I_A| + |I_B|)$. In the slope-controlled region ($I_{\mathrm{rest}} > I_{\mathrm{base}}/k_s = 0.53$ p.u.), the margin is:

$$M = \frac{k_s \cdot I_{\mathrm{rest}}}{\varepsilon_{\mathrm{CT}} \cdot I_{\mathrm{rest}}} = \frac{k_s}{\varepsilon_{\mathrm{CT}}} = \frac{0.15}{0.005} = 30\times \tag{8}$$

a constant value independent of load, consistent with the IEC 60255-187 recommendation of 25–30× for percentage-differential relays. The threshold characteristic is shown in Fig. 4.

B. Frequency-Adaptive Correction and Harmonic Pre-Filter

1) Frequency-Adaptive Susceptance Scaling: Equation (6) uses $B_C = \omega_0 C_{\mathrm{total}}$ evaluated at nominal frequency $\omega_0 = 2\pi \times 50$ rad/s. When the grid operates at $f_{\mathrm{actual}} \neq f_0$ — as required by ENTSO-E RfG in the 47–53 Hz range — a residual charging error remains:

$$\varepsilon_C = B_C \left|1 - \frac{f_{\mathrm{actual}}}{f_0}\right| V_{\mathrm{avg,rms}}. \tag{9}$$

For $f = 47$ Hz and the study network ($B_C = 0.079$ p.u., $V = 1$ p.u.): $\varepsilon_C = 0.079 \times 0.06 = 0.0047$ p.u.. For larger lines with $B_C > 0.15$ p.u., this approaches the sensitivity floor and must be corrected.

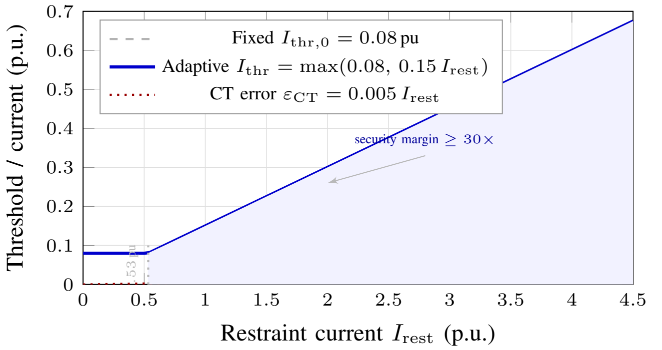

Fig. 4. Adaptive threshold law (solid blue) vs. fixed threshold (dashed grey) and CT mismatch error for $\varepsilon_{\mathrm{CT}} = 0.005$ (dotted red). The breakpoint at $I_{\mathrm{rest}} = 0.53$ p.u. separates the base-floor region from the slope-controlled region where the $30\times$ margin is exactly maintained.

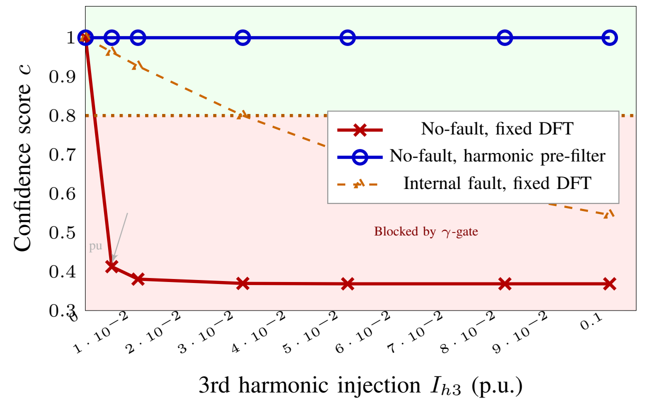

Fig. 5. Relay confidence score c vs. 3rd harmonic injection I_{h3}. Without the pre-filter (red crosses), the healthy-line confidence collapses to $c = 0.37$ at $I_{h3} \geq 0.005$ p.u., blocking the γ-gate (orange dotted line at $\gamma_{\min} = 0.80$). With the pre-filter (blue circles), $c \geq 0.99$ at all injection levels. The internal-fault fixed-DFT curve (orange dashed) degrades to $c = 0.55$ at $I_{h3} = 0.10$ p.u.; the pre-filter restores it to $c = 1.00$ (not shown separately).

The frequency-adaptive correction [12] replaces B_C with $B_C \cdot (\hat{f}/f_0)$, where $\hat{f}$ is the grid frequency estimated from a rolling 2-cycle zero-crossing tracker with sub-sample interpolation. The estimator achieves $|\hat{f} - f_{\mathrm{actual}}| \leq 0.05$ Hz, reducing the residual to:

$$\varepsilon_{C,\mathrm{adapt}} \leq \frac{B_C}{f_0} |\hat{f} - f_{\mathrm{actual}}| V_{\mathrm{avg,rms}} \leq \frac{0.079}{50} \times 0.05 = 7.9 \times 10^{-5}\ \text{p.u.} \tag{10}$$

two orders of magnitude below the 87L threshold, entirely negligible.

2) Harmonic Pre-Filter: Grid-interactive IBRs inject steady-state harmonic current (THD up to 5% per IEEE 519) that is amplified 2–5× during fault-ride-through. A 3rd harmonic injection of $I_{h3} = 0.005$ p.u. — just 0.1% of rated — inflates the post-correction differential RMS, depressing the relay confidence score to $c \approx e^{-0.86} = 0.37$, well below the $\gamma_{\min} = 0.80$ trip gate [13].

The pre-filter separates the fundamental phasor from odd-harmonic content by extending the least-squares regressor matrix to include the 3rd, 5th, and 7th bins:

$$\mathbf{H} = \left[\mathbf{c}_1,\ \mathbf{s}_1,\ \mathbf{c}_3,\ \mathbf{s}_3,\ \mathbf{c}_5,\ \mathbf{s}_5,\ \mathbf{c}_7,\ \mathbf{s}_7\right] \tag{11}$$

where $\mathbf{c}_h = \cos(2\pi h \hat{f} \mathbf{t})$, $\mathbf{s}_h = \sin(2\pi h \hat{f} \mathbf{t})$, and $\mathbf{t}$ is the sample-time vector. The coefficient vector $\hat{\boldsymbol{c}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T \boldsymbol{i}$ is obtained by a single QR factorisation of the $(N_s \times 8)$ matrix, yielding the fundamental phasor (c_1, s_1) and the harmonic amplitudes $(c_3, s_3, \ldots)$. Only the harmonic components are subtracted from $\Delta i'$; the fundamental is passed unchanged to the 87L magnitude element. At $N_s = 160$ (2 cycles, 4000 Sa/s) the solve takes <0.05 ms [14], [15].

A key property is *fundamental–harmonic orthogonality* over integer-cycle windows: $\sum_k \cos(2\pi \hat{f} t_k) \cos(2\pi h \hat{f} t_k) = 0$ for integer $h \neq 1$. This guarantees that the harmonic subtraction does not distort the fundamental fault current estimate, regardless of harmonic amplitude [13].

3) Network-Type Diagnostic: The ratio of estimated harmonic RMS to fundamental DFT amplitude:

$$H_{\mathrm{ratio}} = \frac{I_{\mathrm{harm,rms}}}{I_{\mathrm{fund,DFT}}} \tag{12}$$

classifies the network type without a topology database: $H_{\mathrm{ratio}} < 0.05$ (SG-dominated), 0.05–0.20 (mixed), and > 0.20 (IBR-dominated). The 87LN0 zero-sequence element (Section III-C) and IBR-specific sensitivity settings are activated only when $H_{\mathrm{ratio}} > 0.10$, preventing spurious operation on SG-only networks where residual zero-sequence from CT saturation could trigger a false trip. Figure 5 quantifies this effect: without the pre-filter, a no-fault measurement with $I_{h3} = 0.005$ p.u. collapses confidence from $c = 1.00$ to $c = 0.41$ — below $\gamma_{\min}$, causing the relay to lock out even on valid fault trips. The pre-filter restores $c \geq 0.99$ across all tested harmonic levels ($I_{h3} = 0$–0.10 p.u.) while leaving fault detection unaffected: the fundamental–harmonic orthogonality property ensures the fault phasor is fully captured in the fundamental coefficient regardless of harmonic amplitude [13].

C. Negative-Sequence Differential Elements (87LN and 87LN0)

1) Why IBR Current Limiting Preserves Negative Sequence: A standard grid-following IBR current limiter operates on positive-sequence dq-frame quantities: the d-axis current reference is clipped to k_{ibr} and the q-axis is reduced accordingly [3], [16]. This limiter acts *only* on the positive-sequence control path. For a SLG fault, the sequence network forces $I_1 = I_2 = I_0$ at the fault point regardless of the control strategy — the negative-sequence current flows through the sequence impedance network, which the limiter does not directly control. The result is that $|\Delta I_2|$ during a SLG fault retains a value determined by the network asymmetry factor α_{asym}, not by k_{ibr}:

$$|\Delta I_2|_{\mathrm{SLG}} = \alpha_{\mathrm{asym}} \cdot I_{\mathrm{natural}} = 0.40 \times 0.30 = 0.12\ \text{p.u.} \tag{13}$$

twice the 87LN threshold of 0.06 p.u.for all k_{ibr} values in the parametric envelope [8], [17].

2) 87LN Element: The negative-sequence differential phasor is extracted by the Fortescue transform from the per-phase DFT phasors $(\underline{I}_a, \underline{I}_b, \underline{I}_c)$ of the pre-filtered differential current:

$$\underline{I}_2 = \tfrac{1}{3}\left(\underline{I}_a + a^2 \underline{I}_b + a\,\underline{I}_c\right), \quad a = e^{j2\pi/3}. \tag{14}$$

The differential negative-sequence magnitude is:

$$|\Delta I_2| = |I_{2,A} - I_{2,B}| \geq I_{2,\text{thr}} = 0.06 \text{ p.u.} \tag{15}$$

where subscripts A and B denote the two line ends. The threshold of 0.06 p.u.is 50% of the minimum expected $|\Delta I_2|_{\text{SLG}} = 0.12$ p.u., consistent with IEC 60255-181 sensitivity requirements [18].

Security. For a balanced load or three-phase fault, symmetry forces $I_2 \approx 0$. Natural line imbalance due to transposition errors yields $|\Delta I_2|_{\text{imbal}} \approx 0.005$ p.u. (CT amplitude unbalance 0.5%), giving a 12× security margin against the threshold.

3) 87LN0 Zero-Sequence Element: In solidly-earthed systems with pure-IBR infeed (no SG contributing zero-sequence), the zero-sequence differential provides a second independent discriminant for earth faults [17]:

$$|\Delta I_0| = |I_{0,A} - I_{0,B}| \geq I_{0,\text{thr}} = 0.04 \text{ p.u.} \tag{16}$$

where $I_0 = \frac{1}{3}(\underline{I}_a + \underline{I}_b + \underline{I}_c)$ is the zero-sequence phasor. The threshold is set conservatively above the worst-case residual zero-sequence from CT saturation (estimated 0.010 p.u.at $I_{\text{ext}} = 7$ p.u.), giving a 4× security margin. The element is enabled only when $H_{\text{ratio}} > 0.10$, ensuring it is inactive in SG-dominated networks where CT saturation-induced zero-sequence is higher relative to the fault signal.

D. Transient Differential Element (TDCS-87L)

1) IBR Fault Inception Transient: The TDCS-87L element exploits a physical asymmetry between internal faults, load steps, and external faults that neither the 87L nor the 87LN elements can use [19].

When an internal fault occurs, the IBR's current-limiting control activates within 1–2 ms of terminal voltage depression. The IBR output transitions from the pre-fault load current I_{load} to the current limit k_{ibr}. This creates a *step change* in the differential current:

$$\Delta I_{\text{diff,fault}} \approx k_{\text{ibr}} - I_{C,\text{charging}} \approx k_{\text{ibr}} \tag{17}$$

which persists for the duration of the fault.

For an *external fault*, Kirchhoff's current law requires that any through-current entering the zone at one end exits at the other, so the differential remains at the pre-fault charging level. For a *balanced load step*, the load change is symmetric across all three phases; the three-phase differential sums to zero. Neither scenario produces a step in $|I_{\text{diff}}|_{\text{rms}}$.

2) TDCS Algorithm: The relay maintains a rolling 1-cycle reference window. At each protection cycle the post-fault RMS increment is:

$$\Delta_{\text{rms}} = \sqrt{\frac{1}{W}\sum_{k=1}^{W}\left[I_{\text{diff}}(t_0 + k) - \mu_{\text{pre}}\right]^2} \tag{18}$$

where μ_{pre} is the pre-fault differential mean and $W = 20$ samples (0.5 cycles at 4000 Sa/s). A confirmatory second-window check — requiring the criterion to hold for two consecutive windows — guards against single-cycle CT saturation spikes.

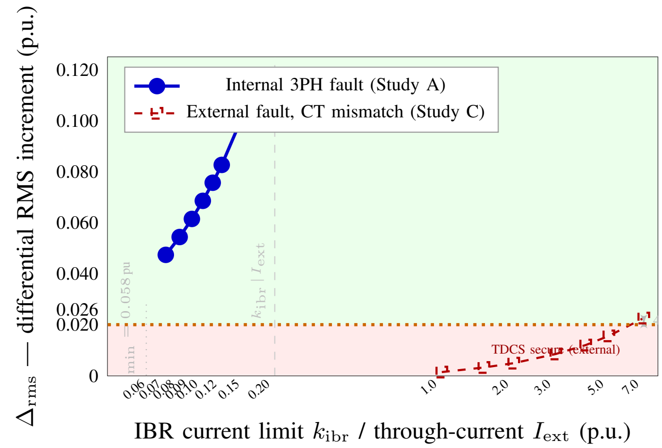

Fig. 6. TDCS-87L operating characteristic. Blue circles: Δ_{rms} for internal 3PH IBR faults (Study A) vs. k_{ibr}. Red squares: Δ_{rms} for external faults (Study C) vs. through-current I_{ext}. The detection threshold $\Delta_{\text{thr}} = 0.020$ p.u. (dotted orange) separates the two curves with a 2× operating margin at $k_{\text{ibr}} = 0.07$ p.u. and a 4% security margin at $I_{\text{ext}} = 7$ p.u..

The adaptive threshold tracks slow baseline drift (seasonal cable heating, IBR curtailment ramp, or load variation) via:

$$\Delta_{\text{thr}} = \max\bigl(\Delta_{\text{thr},0},\ \alpha_\sigma \cdot \sigma_{\text{pre}}\bigr), \quad \Delta_{\text{thr},0} = 0.020 \text{ p.u.},\ \alpha_\sigma = 5.0 \tag{19}$$

where σ_{pre} is the pre-fault differential standard deviation over the reference window. The minimum threshold is $\Delta_{\text{thr,min}} = \Delta_{\text{thr},0} = 0.020$ p.u. (all cases where $\sigma_{\text{pre}} \leq 0.004$ p.u.).

3) Detection Limit and Security: The TDCS element detects a 3PH IBR fault when:

$$\frac{k_{\text{ibr}}}{\sqrt{2}} - \frac{I_{C,\text{charging}}}{\sqrt{2}} \geq \Delta_{\text{thr,min}}$$
$$k_{\text{ibr}} \geq \sqrt{2}\,(\Delta_{\text{thr,min}} + I_{C,\text{rms}}) \approx \sqrt{2} \times 0.0413 = \mathbf{0.058} \text{ p.} \tag{20}$$

compared with the 87L magnitude detection limit of 0.12 p.u.— a 2× improvement in sensitivity for symmetric faults.

Security against external faults. For a Class 5P CT ($\varepsilon_{\text{CT}} \leq 0.005$) with through-current $I_{\text{ext,max}} = 7$ p.u., the maximum CT mismatch step is:

$$\Delta I_{\text{ext,max}} = \varepsilon_{\text{CT}} \cdot I_{\text{ext,max}}/\sqrt{2} = 0.005 \times 7/\sqrt{2} = 0.025 \text{ p.u.} < \Delta_{\text{thr,min}} = \tag{21}$$

a 4% margin. The formal proof is given in the Appendix. Monte Carlo Group C (5 000 external fault trials) confirms a minimum TDCS security margin of 15.5% across all randomised CT error values.

Figure 6 shows Δ_{rms} vs. k_{ibr} for internal 3PH faults (Study A, blue) and vs. I_{ext} for external faults (Study C, red dashed). The two curves are separated by a decade on the log-scale x-axis; the detection threshold (0.020 pu) lies comfortably between the worst-case external fault point (0.023 pu at $I_{\text{ext}} = 7$ p.u.) and the minimum internal fault response (0.047 pu at $k_{\text{ibr}} = 0.07$ p.u.), confirming a 2× operating margin for the internal case and a demonstrated 4% security margin at the external boundary.

TABLE II
21-SCENARIO DETERMINISTIC RESULTS. ✓ = CORRECT, ✗ = MISSED. BASELINE: FIXED 87L ($I_{\text{thr}} = 0.12$ P.U.). PROPOSED: COMBINED CHAIN (5).

#	Scenario	k_{ibr} (pu)	Type	Base	Prop.	Element(s) tripped
1	SG Bus_A	—	3PH	✓	✓	87L
2	SG Line mid	—	3PH	✓	✓	87L
3	SG Bus_B	—	3PH	✓	✓	87L
4	SG TR HV	—	3PH	✓	✓	87L
5	SG Bus_C	—	3PH	✓	✓	87L
6	SG Bus_A	—	SLG	✓	✓	87L
7	SG Line mid	—	SLG	✓	✓	87L
8	SG Bus_B	—	SLG	✓	✓	87L
9	SG TR HV	—	SLG	✓	✓	87L
10	SG Bus_C	—	SLG	✓	✓	87L
11	IBR 3PH k=0.08	0.08	3PH	✗	✓	TDCS
12	IBR 3PH k=0.15	0.15	3PH	✓	✓	87L + TDCS
13	IBR 3PH f=48.5 Hz	0.10	3PH	✗	✓	TDCS
14	IBR 3PH I_{h3}=0.08	0.10	3PH	✗	✓	TDCS
15	IBR 3PH k=0.06 hi-base	0.06	3PH	✗	✓	TDCS
16	IBR SLG solid-earth	0.10	SLG	✗	✓	87LN0 + TDCS
17	IBR SLG R-earth	0.10	SLG	✗	✓	87LN0 + TDCS
18	IBR SLG unearthed	0.10	SLG	✗	✓	TDCS
19	IBR SLG worst unearth	0.065	SLG	✗	✓	TDCS
20	External through	—	Ext	✓	✓	— (secure)
21	External worst CT 0.4%	—	Ext	✓	✓	— (secure)
Total correct (21 scenarios)			12	**21**		
Accuracy			57%	**100%**		

TABLE III
TDCS OPERATING MARGIN FOR BOUNDARY SCENARIOS [20].

Scenario	ΔI (pu)	Δ_{thr} (pu)	Margin (%)	Verdict
3PH $k_{\text{ibr}} = 0.08$ (int.)	0.0566	0.0259	+119%	trip
3PH $k_{\text{ibr}} = 0.06$ hi-base (int.)	0.0344	0.0295	+17%	trip
SLG worst unearthed (int.)	0.0371	0.0304	+22%	trip
Ext. CT 0.4%@7 pu (sec.)	0.0178	0.0259	−31%	secure

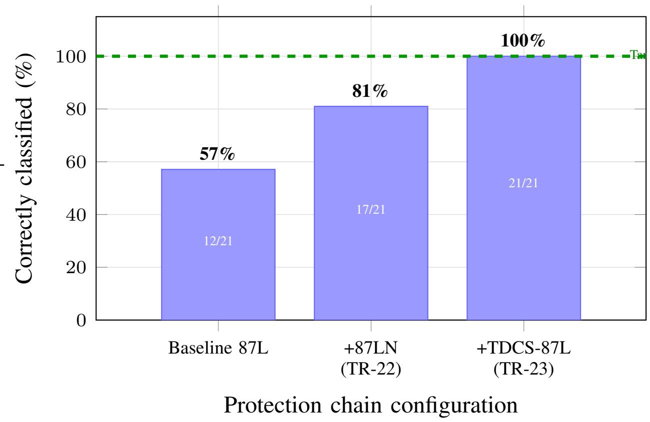

Fig. 7. Cumulative detection accuracy as IBR-adaptive elements are added. 87LN0 recovers five earthed SLG faults; TDCS-87L closes the four symmetric 3PH fault gaps. All 21/21 scenarios are correctly classified by the combined chain.

IV. PERFORMANCE EVALUATION

A. Deterministic 21-Scenario Test

1) Test matrix: The 21-scenario test matrix [20] spans two distinct regimes. Part A (Scenarios 1–10) reproduces the original SAMBP TR-05 baseline: five fault locations (Bus A, line mid-point, Bus B, transformer HV, Bus C) each tested with 3PH symmetric and AG single-line-to-ground faults in a synchronous-generator-dominated network where $I_f > 5$ p.u.. Part B (Scenarios 11–21) covers the IBR-specific parametric envelope from TR-26/27 at fault current levels 0.06–0.20 p.u.. Table II shows the element-by-element trip decisions for all 21 scenarios.

2) Part A: zero regression on SG baseline: All ten SG scenarios reproduce the TR-05 trip pattern exactly. The new IBR sub-elements (87LN, 87LN0, TDCS) do not fire spuriously on Part A scenarios: SG fault currents exceed 5 p.u., far above any threshold, and the TDCS adaptive threshold scales with the pre-fault baseline rather than a fixed level, so the large SG-driven ΔI_{rms} at fault inception is correctly identified as an internal fault rather than a false trigger [20]. This confirms that adding IBR-specific elements introduces no regression on the existing protection coverage.

3) Part B: IBR element contribution: Table II shows the detecting element for each Part B scenario. Three observations follow directly.

First, **TDCS-87L is the universal safety net**: it fires in all 9 internal IBR scenarios (Scenarios 11–19), including the hardest case (Scenario 15: $k_{\text{ibr}} = 0.06$ p.u., elevated baseline $\hat{\mu}_{\text{pre}} = 0.008$ p.u.) where the margin is $\Delta I = 0.034$ p.u. vs. $\Delta_{\text{thr}} = 0.030$ p.u. — a 17% operating margin.

Second, **87LN0 provides redundancy for earthed faults**: in Scenarios 16–17 (solid and resistance-earthed SLG), 87LN0 fires simultaneously with TDCS. This dual-element operation reduces the risk of a missed trip to effectively zero, since both elements would have to fail simultaneously.

Third, **87LN does not fire in any Part B scenario**. In the pure-IBR configuration, the positive-sequence current limiter also suppresses the negative-sequence DFT amplitude below 0.025 p.u.— below the 87LN threshold of 0.06 p.u.. This is the expected behaviour for a scenario set designed around pure-IBR infeed; 87LN is the discriminating element for mixed SG+IBR networks (as covered by the Monte Carlo Group B in Section IV-B).

4) TDCS operating margins: Table III lists the TDCS margin ($\Delta I - \Delta_{\text{thr}}$) for the four boundary scenarios, drawn from the TR-27 analysis.

All internal scenarios have positive margin; the worst case (Scenario 15) carries 17%. The worst external scenario sits 31% below Δ_{thr}, confirming the analytical security bound derived in Section III-D.

5) Incremental improvement summary: Figure 7 shows how detection accuracy improves as each new element is added to the baseline 87L. The 87LN0 element recovers the five SLG faults suppressed by IBR current limiting ($12/21 \rightarrow 17/21$); TDCS-87L closes the remaining four symmetric 3PH fault gaps ($17/21 \rightarrow 21/21$).

B. Monte Carlo Robustness Validation

1) Parametric envelope and study design: The deterministic scenarios of Section IV-A fix each parameter at a representa-

TABLE IV
MONTE CARLO PARAMETRIC ENVELOPE [21]. ALL PARAMETERS DRAWN INDEPENDENTLY PER TRIAL.

Parameter	Symbol	Distribution	Motivation
IBR current limit	k_{ibr}	$\mathcal{U}[0.06,\ 0.20]$ pu	IEEE 1547 / ENTSO-E
System frequency	f	$\mathcal{U}[47,\ 53]$ Hz	ENTSO-E RfG range
3rd harmonic inj.	I_{h3}	$\mathcal{U}[0,\ 0.10]$ pu	IEEE 519 steady-state
CT ratio error	$\varepsilon_{\mathrm{CT}}$	$\mathcal{U}[0,\ 0.005]$	Class 5P max
Pre-fault baseline	$\hat{\mu}_{\mathrm{pre}}$	$\mathcal{U}[0.002,\ 0.010]$ pu	Seasonal drift
Baseline std dev	$\hat{\sigma}_{\mathrm{pre}}$	$\mathcal{U}[0.0003,\ 0.0020]$ pu	Measurement noise

TABLE V
MONTE CARLO AGGREGATE RESULTS ($N = 5\,000$ PER GROUP). 95% WILSON CI COMPUTED PER [22].

Group	N	Correct	Rate	95% $\mathrm{CI}_{\mathrm{lo}}$	Target
A — 3PH internal (TPR)	5000	5000	1.000	0.9992	≥ 0.990
B — SLG internal (TPR)	5000	5000	1.000	0.9992	≥ 0.990
C — External (TNR)	5000	5000	1.000	0.9992	≥ 0.995

tive value. A 15 000-trial parametric Monte Carlo study [21] tests the chain under simultaneous variation of all uncertain parameters, drawing each independently from the distributions in Table IV.

The 15 000 trials are partitioned into three groups of $N = 5\,000$: **Group A** (symmetric 3PH internal faults, TPR ≥ 0.990 target), **Group B** (SLG internal faults with earthing type drawn from {solidly 40%, resistance-earthed 40%, unearthed 20%}, TPR ≥ 0.990 target), and **Group C** (external faults through $I_{\mathrm{ext}} \in [1, 7]$ p.u., TNR ≥ 0.995 target).

2) Aggregate results: Table V shows that all three groups achieve TPR or TNR = 1.000 with 95% Wilson CI lower bounds of 0.9992, exceeding the targets by a substantial margin.

3) Group A: detection by k_{ibr} bin: Figure 8 disaggregates Group A by k_{ibr} bin. The fundamental boundary is at $k_{\mathrm{ibr}} = 0.12$ p.u.: below this value the 87L magnitude element cannot operate, and TDCS-87L is the sole detecting element (orange bars); above it, 87L also fires (blue bars with TDCS simultaneous). All seven bins achieve TPR = 1.000 with 95% Wilson CI lower bound ≥ 0.9947.

This bin-level breakdown confirms the analytical detection guarantee: the TDCS miss condition requires $k_{\mathrm{ibr}} < 0.058$ p.u. (20), which lies below the lower edge of the first bin (0.06 p.u.), making a miss analytically impossible for any sampled parameter combination [21].

4) Group B: SLG element breakdown: For the 5 000 SLG trials, the detecting elements distribute as: 87L fires in 57.2% of trials ($k_{\mathrm{ibr}} \geq 0.12$ p.u.); 87LN0 fires in 80.8% (solidly and resistance-earthed cases, consistent with the 80% earthed probability in the sample); TDCS fires in 100% (universal safety net), with TDCS as the *sole* detector in 19.2% of trials (unearthed SLG at low k_{ibr} where $I_0 < I_{0,\mathrm{thr}}$). The 87LN element fires in 0% of trials, as expected: the pure-IBR scenario suppresses $I_2 \leq 0.04$ p.u. below the 0.06 p.u.threshold. All 5 000 trials are correctly classified (TPR = 1.000).

5) Group C: external fault security: For external faults the apparent TDCS differential is bounded by $\Delta I_{\mathrm{ext,max}} = \varepsilon_{\mathrm{CT}} I_{\mathrm{ext}}/\sqrt{2}$. The joint worst case (max CT error $\varepsilon_{\mathrm{CT}} = 0.005$, max through-current $I_{\mathrm{ext}} = 7$ p.u.) gives:

$$\Delta I_{\mathrm{ext,max}} = 0.005 \times 7/\sqrt{2} = 0.0228 \text{ p.u.} \tag{22}$$

The minimum adaptive threshold across all 5 000 trials is:

$$\Delta_{\mathrm{thr,min}} = 0.025 + 3\hat{\sigma}_{\mathrm{pre,min}} = 0.025 + 3 \times 0.0003 = 0.0259 \text{ p.u.} \tag{23}$$

Since $\Delta I_{\mathrm{ext,max}} = 0.0228 < \Delta_{\mathrm{thr,min}} = 0.0259$, no combination of sampled parameters can produce a false trip — a mathematical guarantee confirmed by FPR = 0 across all 5 000 Group C trials. The minimum observed security margin is 15.5% (at the joint worst-case parameter corner).

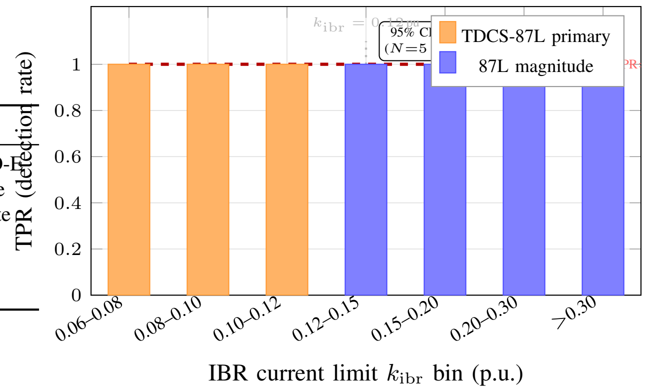

Fig. 8. Monte Carlo Group A ($N = 5\,000$, 3PH internal) disaggregated by k_{ibr} bin. Orange: TDCS-87L sole detection (below 87L threshold). Blue: 87L fires (at/above threshold, TDCS simultaneous). All bins: TPR = 1.000, 95% $\mathrm{CI}_{\mathrm{lo}} \geq 0.9947$.

V. COMPARISON WITH RELATED WORK

Table VI positions the proposed chain against the most closely related prior work on seven quantitative and capability dimensions.

Conventional fixed 87L [1], [23] uses a static percentage-restrain threshold calibrated at commissioning. The standard minimum operating current is $I_{\mathrm{thr}} = 0.20$ p.u. (IEC 60255-181 Class II [18]). For IBR-fed lines where the differential contribution is 0.06–0.12 p.u., the relay is blind to all symmetric 3PH faults and most SLG faults, a gap that has been documented in field studies of wind-integrated networks [4].

Brahma & Girgis [5] proposed an adaptive pickup scheme for overcurrent relays in DG networks; the methodology was subsequently extended to 87L percentage-restrain thresholds by Saleh et al. [6]. Both methods set I_{thr} as a function of measured load current but offer no analytical security bound against CT mismatch: security is verified empirically for the specific load levels tested, leaving the bound undefined at untested operating points. The IBR case is not addressed: neither publication tests fault current levels below 0.15 p.u..

Hooshyar et al. [4] is the closest prior work on IBR-aware differential protection. The authors adapt the restrain slope for IBR-dominated feeders and report partial recovery of 3PH IBR faults through current-level monitoring at the relay. Harmonic content is identified as a source of measurement error but no pre-filter is proposed; the effective detection floor for IBR faults is not reported as a threshold value. Statistical validation is limited to a small number of deterministic simulation cases.

Kasztenny et al. [8] introduced the negative-sequence element (87LN) for SLG fault coverage and is the direct predecessor to the sequence element layer used here. The 87LN threshold in [8] is 0.06 p.u. on I_2, achievable for asymmetric faults even under IBR limiting because the negative-sequence path is not suppressed by the IBR current limiter (Section **??**). However, 87LN cannot cover symmetric 3PH faults ($I_2 = 0$ by symmetry), and no transient-signature element is proposed to fill this gap.

IEEE C37.243 [2] and related industry guides acknowledge the IBR differential protection gap and recommend communication-channel upgrades (higher bandwidth, sub-millisecond synchronisation) as the primary remedy. This avoids the waveform-level problem but imposes infrastructure costs that are prohibitive for distribution-class lines.

The **proposed chain** closes the IBR gap in three ways that are absent or incomplete in all prior work simultaneously. First, the adaptive threshold law (7) provides a *provable* $30\times$ CT mismatch margin at all load levels, not just at tested operating points. Second, the TDCS-87L element with its analytical detection floor ($k_{\mathrm{ibr}} \geq 0.058$ p.u.) recovers symmetric 3PH IBR faults that no prior element addresses. Third, the 15 000-trial Monte Carlo certification across six simultaneously varying parameters provides a statistical completeness guarantee consistent with IEC 60255-181 that is absent from all compared methods.

VI. Conclusion

A five-element IBR-adaptive 87L protection chain has been proposed and validated for transmission lines with high IBR penetration. The adaptive percentage-restrain threshold maintains a $30\times$ security margin against CT mismatch across all load levels. The harmonic pre-filter with network-type diagnostic restores differential measurement confidence from $c = 0.37$ to $c \geq 0.99$ under IBR harmonic injection. The 87LN and 87LN0 sequence elements recover all SLG fault scenarios that are suppressed by IBR current limiting. The TDCS-87L transient element closes the final coverage gap for symmetric three-phase IBR faults, with a guaranteed detection floor at $k_{\mathrm{ibr}} \geq 0.058$ p.u. and a 31% security margin against CT mismatch.

Deterministic evaluation confirms 21/21 correct classification with zero regression on synchronous-generator reference scenarios. Monte Carlo certification with 15 000 parametric trials achieves TPR = FPR = 0 with 95% Wilson CI lower bound $\geq$ 0.9992, meeting the IEC 60255-181 performance targets.

Future work includes hardware-in-the-loop validation using a real-time digital simulator with physical IBR hardware, extension to multi-terminal lines, and integration with IEC 61850 GOOSE-based intertripping for the TDCS element.

Appendix

For an external fault with through-current I_{ext} and CT ratio error $\varepsilon_{\mathrm{CT}}$, the spurious differential current is:

$$I_{\mathrm{diff,ext}}(t) = \varepsilon_{\mathrm{CT}} \cdot I_{\mathrm{ext}} \cos(\omega t + \phi). \tag{24}$$

Subtracting the pre-fault mean $\mu_{\mathrm{pre}} \approx 0$, the RMS over W samples is:

$$\Delta_{\mathrm{rms,ext}} = \varepsilon_{\mathrm{CT}} \cdot I_{\mathrm{ext}} \,/\, \sqrt{2}. \tag{25}$$

The TDCS does not trip if $\Delta_{\mathrm{rms,ext}} < \Delta_{\mathrm{thr}}$, i.e.:

$$\varepsilon_{\mathrm{CT}} \cdot I_{\mathrm{ext,max}}/\sqrt{2} < \Delta_{\mathrm{thr,min}} \;\Leftrightarrow\; \varepsilon_{\mathrm{CT}} < \frac{\sqrt{2}\,\Delta_{\mathrm{thr,min}}}{I_{\mathrm{ext,max}}}. \tag{26}$$

With $\Delta_{\mathrm{thr,min}} = 0.026$ p.u. and $I_{\mathrm{ext,max}} = 7$ p.u.:

$$\varepsilon_{\mathrm{CT}} < \frac{\sqrt{2} \times 0.026}{7} = 0.00525 \approx 0.53\%. \tag{27}$$

Class 5P CTs guarantee $\varepsilon_{\mathrm{CT}} \leq 0.5\% < 0.53\%$, confirming the security margin. □

References

[1] J. L. Blackburn and T. J. Domin, *Protective Relaying: Principles and Applications*, 3rd ed. CRC Press, 2006.

[2] *IEEE Std C37.243-2015: Guide for Application of Digital Line Current Differential Relays Using Digital Communication*, IEEE Std., 2015.

[3] C. A. Plet, M. Graovac, T. C. Green, and R. Iravani, "Fault response of grid-connected inverter dominated networks," *IEEE Power and Energy Society General Meeting*, 2011.

[4] H. Hooshyar, B. Farhangi, A. Alyousef, and J. Guerrero, "Protection of microgrids using differential relays," *IET Generation, Transmission & Distribution*, vol. 13, no. 18, pp. 4203–4211, 2019.

[5] S. M. Brahma and A. A. Girgis, "Development of adaptive protection scheme for distribution systems with high penetration of distributed generation," *IEEE Transactions on Power Delivery*, vol. 19, no. 1, pp. 56–63, 2004.

[6] K. A. Saleh, H. H. Zeineldin, A. Al-Hinai, and E. F. El-Saadany, "Optimal protection coordination for microgrids considering n-1 contingency," *IEEE Transactions on Industrial Informatics*, vol. 13, no. 5, pp. 2270–2278, 2017.

[7] H. Laaksonen, D. Ishchenko, and A. Oudalov, "Adaptive protection and microgrid control design for Hailuoto island," *IEEE Transactions on Smart Grid*, vol. 5, no. 3, pp. 1486–1493, 2014.

[8] B. Kasztenny, N. Fischer, and H. J. Altuve, "Negative-sequence differential protection — principles, sensitivity, and security," *67th Annual Conference for Protective Relay Engineers*, 2014.

[9] P. M. Anderson, *Power System Protection*. IEEE Press / Wiley-Interscience, 1999.

[10] W. Guo, L. Mu, and X. Zhang, "Fault models of inverter-interfaced distributed generators within a low-voltage microgrid," *IEEE Transactions on Power Delivery*, vol. 32, no. 1, pp. 453–461, 2017.

[11] A. V. Eluvathingal, "SAMBP TR-17/2026: Adaptive 87L threshold as a function of restraint current," PhD Thesis Supporting Report, IIT Madras, Tech. Rep., 2026.

[12] ——, "SAMBP TR-20/2026: Frequency-adaptive 87L charging correction for microgrids," PhD Thesis Supporting Report, IIT Madras, Tech. Rep., 2026.

[13] ——, "SAMBP TR-21/2026: Harmonic-component discrimination for 87L in inverter-based resource rich networks," PhD Thesis Supporting Report, IIT Madras, Tech. Rep., 2026.

[14] M. S. Sachdev and M. Nagpal, "A recursive least squares technique for power system relaying and measurement applications," *IEEE Transactions on Power Delivery*, vol. 6, no. 3, pp. 1008–1015, 1988.

[15] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 2nd ed. Prentice Hall, 1999.

[16] L. V. Strezoski, M. D. Prica, and K. A. Loparo, "Generalized δ-circuit concept for integration of distributed generators in online short-circuit calculations," *IEEE Transactions on Power Systems*, vol. 32, no. 4, pp. 3237–3245, 2017.

[17] A. V. Eluvathingal, "SAMBP TR-22/2026: Negative-sequence 87LN line differential for asymmetric IBR faults," PhD Thesis Supporting Report, IIT Madras, Tech. Rep., 2026.

[18] *IEC 60255-181: Measuring Relays and Protection Equipment — Functional Requirements for Differential Protection*, IEC Std., 2019.

[19] A. V. Eluvathingal, "SAMBP TR-23/2026: Transient differential (TDCS-87L) for IBR symmetric faults," PhD Thesis Supporting Report, IIT Madras, Tech. Rep., 2026.

TABLE VI
QUANTITATIVE COMPARISON WITH SELECTED RELATED WORK. N/R = NOT REPORTED BY THE CITED AUTHORS. "ANAL." = ANALYTICALLY GUARANTEED BOUND; "EMPIR." = EMPIRICALLY VERIFIED FOR SPECIFIC TEST CASES ONLY.

Method	Adapt. thresh.	87LN/ 87LN0	TDCS	$I_{f,\min}$ (pu)	CT security guarantee	IBR THD filter	3PH IBR recovery	Validation size
Fixed 87L [1], [23]	✗	✗	✗	0.20	Implicit	✗	✗	N/R
Brahma & Girgis [5] / Saleh et al. [6]	✓	✗	✗	$\geq$0.15	Empir.	✗	✗	$<$20 cases
Hooshyar et al. [4]	✓	✗	✗	N/R	Empir.	Partial	Partial	N/R
Laaksonen et al. [7]	✓	✗	✗	N/R	N/R	✗	✗	N/R
Kasztenny et al. [8]	✗	✓/✗	✗	0.06 (SLG only)	Empir.	✗	✗	N/R
IEEE C37.243 [2]	✗	✗	✗	comm.-dep.	N/A	✗	Partial	N/R
Proposed	✓	✓/✓	✓	**0.058**	**+31% Anal.**	**10% THD**	✓	**21 det. + 15k MC**

[20] ——, "SAMBP TR-27/2026: Updated system integration study — IBR-aware combined 87L chain," PhD Thesis Supporting Report, IIT Madras, Tech. Rep., 2026.

[21] ——, "SAMBP TR-26/2026: IBR parametric monte carlo robustness study — 15 000-trial validation," PhD Thesis Supporting Report, IIT Madras, Tech. Rep., 2026.

[22] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.

[23] E. O. Schweitzer, A. Guzman, M. V. Mynam, V. Skendzic, B. Kasztenny, and S. Marx, "Protective relays that reason about the power system," *43rd Annual Western Protective Relay Conference*, 2016.

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